

SIE 433/533 Fundamentals of Data Science for Engineer

Homework 3

Due: March 20 (11:59PM), 2025

Instruction:

There are **3 (three) problems** in total. Each of them may contain several sub-problems.

If you want to copy&paste your code and highlight the code, (1) go to <https://highlight.hohli.com/> (2) choose Python as the language (3) copy your code to the console, and (4) click *Highlight*.

Submission Guidelines:

- Ensure to compile all your solutions into a **single PDF file** and submit it via D2L before the deadline.
- **Detailed intermediate steps are required for scoring.** Solutions that present only the final answer without supporting work will not be awarded any points.
- For problems solved using Python, **include the corresponding code within your PDF submission.** To aid in evaluating your work, please highlight the key segments of your code. This will streamline the review process, particularly in identifying any errors in your solution approach.

Problem 1: (4 points)

Diabetes model selection using AIC, BIC. In the script provided for the class, i.e. “**Model-Selection.ipynb**”, we applied AIC and BIC information criteria to perform the model selection for the Diabetes dataset using the features {S1, S2, S3, S5}. In this problem, you are asked to apply AIC, BIC for the model selection with the features $\mathbf{X}$, where $\mathbf{X}$ includes {Age, Body mass index, Average blood pressure, S4, S5, S6}, and y is the “target”

- 1) Generate the *AIC value - Degree of Freedom* figure for model selection. To answer this question, you are expected to fit linear regression models for all combinations of the variables in $\mathbf{X}$ to the target and calculate corresponding AICs. (1 points)
 - a. Copy your figure in the answer.
 - b. Report the lowest AIC value and the corresponding variables that are used.
- 2) Generate the *BIC value - Degree of Freedom* figure for model selection. To answer this question, you are expected to fit linear regression models for all combinations of the variables in $\mathbf{X}$ to the target and calculate corresponding BICs. (1 points)
 - a. Copy your figure in the answer.
 - b. Report the lowest BIC value and the corresponding variables that are used.
- 3) Based on the AIC and BIC in previous questions, (2 points)
 - a. Do AIC and BIC select the same model?
 - b. If the **number of samples** in $\mathbf{X}$ increases to 1000000, which one (AIC/BIC) should be used to select a simpler model?

Problem 2: (2 points)

The *prostate* cancer data set contains eight predictors (columns 1-8), $\mathbf{X} \in \mathbb{R}^8$. The outcome variable Y is given by column 9. The last column (column 10) is the train/test indicator, indicating 67 “training” data observations and 30 “testing” observations. Fit the LASSO regression for the *prostate* cancer data set. Select the parameter with 5-fold CV, using the minimum CV rule. Report the best λ , the selected model, the estimated regression coefficients, and the TestErr. (2 points)

(Note: You can use Python functions “*LassoCV()*”)

Question 3 (4 points)

In Page 7 in Lecture 11, we define the mathematical formulation for Logistic regression. In this problem, we want to

- 1) Define the function, i.e., `logistic()` (check the Jupyter Notebook), to generate values for Logistic function using Python. The input of the function is a constant, the output of the function is also a constant, i.e., the function should output 0.5 when the input is 0. Write down the output of the function when the input is 2.7. (2 point)
- 2) Plot the outputs of the function in the interval $[-10, 10]$ by matplotlib at a step size 0.1, i.e., you should plot the dots given the inputs equal to $-10, -9.9, -9.8, \dots, 9.8, 9.9, 10$.
 - a. Please use scatter dots, black color, dots' size is 1. (1 point)
 - b. x label is 'x' and y label is 'y', title of the figure is 'Logistic Regression'. (1 point)

(Note: You should not import the functions such as `np.random.normal()` or any other existing functions that directly calculate the values for the distribution. You are allowed to use `math.sqrt()`, `math.pi`, `math.exp()` or other scientific operators.)